Problem Set 23

Volumes IV: Cylindrical Shells

November 2, 2026

Problem 1

(a)

$$V = \pi h(r_2^2 - r_1^2) = 2\pi \cdot \frac{r_1 + r_2}{2} \cdot h \cdot (r_2 - r_1) = 2\pi r h \Delta r$$

No, the formula would not work if r were the inner radius.

(b)

$$V = 2\pi \cdot 25 \cdot 0.1 \cdot 2.05 = 10.25\pi$$
$$V = 10\pi$$

(c) $2\pi\overline{x}_i$ is the circumference. $f(\overline{x}_i)$ is the height. Δx is the width.

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi\overline{x}_i \, f(\overline{x}_i) \, \Delta x = \int_a^b 2\pi x \, f(x) \, dx$$

Problem 2

(b)

$$y = x - x^2$$
$$V = 2\pi \int_0^1 x^2 - x^3 \, dx$$
$$= 2\pi(\frac{x^3}{3} - \frac{x^4}{4})\Big|_0^1$$
$$= \boxed{\frac{\pi}{6}}$$

(c)

$$V = 2\pi \int_0^1 x(x^3 - 2x^2 + x) \, dx$$
$$= 2\pi(\frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{3})\Big|_0^1$$
$$= \boxed{\frac{\pi}{15}}$$

Problem 3

(b)

$$x = \sqrt[3]{y}$$
$$V = 2\pi \int_0^8 y \cdot \sqrt[3]{y}\, dy$$
$$= 2\pi \cdot \frac{3}{7} y^{\frac{7}{3}} \Big|_0^8$$
$$= \boxed{\frac{768\pi}{7}}$$
$$A = 64 - x^6$$
$$V = \pi \int_0^2 64 - x^6\, dx$$
$$= \pi(64x - \frac{x^7}{7}) \Big|_0^2$$
$$= \boxed{\frac{768\pi}{7}}$$

Problem 4

(b)

$$V = 2\pi \int_0^1 (2 - x)x^4\, dx$$
$$= 2\pi \int_0^1 -x^5 + 2x^4\, dx$$
$$= 2\pi(\frac{2}{5}x^5 - \frac{x^6}{6}) \Big|_0^1$$
$$= \boxed{\frac{7\pi}{15}}$$

(d)

$$x = \sqrt[3]{y}$$
$$V = 2\pi \int_0^1 (1 - x)(1 - \sqrt[3]{x})\, dx$$
$$= 2\pi \int_0^1 x^{\frac{4}{3}} - x - x^{\frac{1}{3}} + 1\, dx$$
$$= 2\pi(\frac{3}{7}x^{\frac{7}{3}} - \frac{x^2}{2} - \frac{3}{4}x^{\frac{4}{3}} + x) \Big|_0^1$$
$$= \boxed{\frac{5\pi}{14}}$$

Problem 5

(a)

$$\begin{aligned} V &= 2\pi \int_1^3 x(x^2 - 4x + 7 - 4(x-2)^2)\,dx \\ &= \begin{bmatrix} u = x-2 \\ du = dx \end{bmatrix} 2\pi \int_{-1}^1 (u+2)(3-3u^2)\,du \\ &= 2\pi \int_{-1}^1 -3u^3 - 6u^2 + 3u + 6\,du \\ &= 2\pi(-\frac{3}{4}u^4 - 2u^3 + \frac{3}{2}u^2 + 6u)\Big|_{-1}^1 \\ &= \boxed{16\pi} \end{aligned}$$

(b)

$$\begin{aligned} V &= 2\pi \int_0^1 xe^{-x^2}\,dx \\ &= \begin{bmatrix} u = x^2 \\ du = 2x\,dx \end{bmatrix} \pi \int_0^1 e^{-u}\,du \\ &= \pi(e^{-u})\Big|_1^0 \\ &= \boxed{\pi(1 - \frac{1}{e})} \end{aligned}$$

The disk method is a different story as it is simply too hard to integrate.

Problem 6

(a)

$$\boxed{V = 2\pi \int_1^2 x \ln x\,dx}$$

(c)

$$\boxed{V = 2\pi \int_0^\pi (4-y)\sqrt{\sin y}\,dy}$$

Problem 7

(a) The solid obtained by rotating the region under $y = x^4$ on $(0, 3)$ about the y-axis.

(b) The solid obtained by rotating the region bounded by $x = 1 - y^2$, $x = 0$, $y = 0$, and $y = 1$ about the line $y = 3$.

Problem 8

(a)

$$x^2 + y^2 = r^2 \implies y = \sqrt{r^2 - x^2}$$
$$V = 4\pi \int_0^r x\sqrt{r^2 - x^2}\, dx$$
$$= \begin{bmatrix} u = r^2 - x^2 \\ du = -2x\, dx \end{bmatrix} - 2\pi \int_{r^2}^0 \sqrt{u}\, du$$
$$= 2\pi \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_0^{r^2}$$
$$= \boxed{\frac{4}{3}\pi r^3}$$

(b)

$$y = -\frac{h}{r}x + h$$
$$V = 2\pi \int_0^r x(-\frac{h}{r}x + h)\, dx$$
$$= 2\pi \int_0^r -\frac{h}{r}x^2 + hx\, dx$$
$$= 2\pi(-\frac{h}{3r}x^3 + \frac{h}{2}x^2)\Big|_0^r$$
$$= 2\pi(\frac{1}{2}r^2h - \frac{1}{3}r^2h)$$
$$= \boxed{\frac{1}{3}\pi r^2 h}$$

(c)

$$x^2 + y^2 = r^2 \implies x = \sqrt{r^2 - \frac{H^2}{4}}$$
$$V = 4\pi \int_{\sqrt{r^2 - \frac{H^2}{4}}}^r x\sqrt{r^2 - x^2}\, dx$$
$$= \begin{bmatrix} u = r^2 - x^2 \\ du = -2x\, dx \end{bmatrix} 2\pi \int_0^{\frac{H^2}{4}} \sqrt{u}\, du$$
$$= 2\pi \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_0^{\frac{H^2}{4}}$$
$$= \boxed{\frac{\pi}{6}H^3}$$

Problem 9

(a)

$$A = \int_0^\pi \sin x \, dx = \boxed{2}$$

(b)

$$V = 2\pi \int_0^\pi x \sin x \, dx$$
$$= \begin{bmatrix} u = x & dv = \sin x \, dx \\ du = dx & v = -\cos x \end{bmatrix} 2\pi x \cos x \Big|_\pi^0 + 2\pi \int_0^\pi \cos x \, dx$$
$$= 2\pi^2 + 2\pi \cdot 0 = \boxed{2\pi^2}$$

(c)

$$V = \pi \int_0^\pi \sin^2 x \, dx$$
$$= \frac{\pi}{2} \int_0^\pi 1 - \cos 2x \, dx$$
$$= \frac{\pi}{2}(x - \frac{\sin 2x}{2})\Big|_0^\pi$$
$$= \boxed{\frac{\pi^2}{2}}$$

(d)

$$V = \boxed{2\pi^2}$$

The region is symmetric about $x = \frac{\pi}{2}$.

(e)

$$V = \pi \int_0^\pi (\sin x + 1)^2 - 1 \, dx$$
$$= \pi \int_0^\pi \sin^2 x + 2 \sin x \, dx$$
$$= \pi \int_0^\pi \frac{1}{2} - \frac{1}{2} \cos 2x + 2 \sin x \, dx$$
$$= \pi(\frac{x}{2} - \frac{\sin 2x}{4} - 2\cos x)\Big|_0^\pi$$
$$= \boxed{\frac{\pi^2}{2} + 4\pi}$$

(f)

$$V = A \cdot 2\pi\overline{x} \implies \overline{x} = \frac{\pi}{2}$$
$$V = A \cdot 2\pi\overline{y} \implies \overline{y} = \frac{\pi}{8}$$
$$\boxed{\left(\frac{\pi}{2}, \frac{\pi}{8}\right)}$$

Problem 10

(a)

$$y = f(x) \implies x = g(y)$$
$$\boxed{V = \int_0^{f(b)} \pi(b^2 - g(y)^2)\,dy}$$

(b)

$$y = f(x) \implies dy = f'(x)\,dx$$
$$V = \int_0^{f(b)} \pi(b^2 - g(f(x))^2)\,dy$$
$$= \pi b^2 f(b) - \int_0^b \pi x^2\,dy$$
$$= \pi b^2 f(b) - \int_0^b \pi x^2 f'(x)\,dx$$
$$\int_0^b \pi x^2 f'(x)\,dx$$
$$= \begin{bmatrix} u = x^2 & dv = f'(x)\,dx \\ du = 2x\,dx & v = f(x) \end{bmatrix} \pi x^2 f(x)\Big|_0^b - 2\pi \int_0^b x f(x)\,dx$$
$$\boxed{V = \int_0^b 2\pi x f(x)\,dx}$$

(c) $\pi b^2 f(b)$ represents the volume of a cylinder with radius b and height $f(b)$. $\int_0^b \pi x^2 f'(x)\,dx$ represents the volume of the solid formed by rotating the region bounded by $y = f(x)$, $x = 0$, and $y = f(b)$ around the y-axis.

(d)

$$V = 2\pi\overline{x}A$$
$$= 2\pi \cdot \frac{1}{A}\int_a^b x f(x)\,dx \cdot A$$
$$= \boxed{\int_a^b 2\pi x f(x)\,dx}$$

(e)

$$f(0) = 0 \implies y = 0$$
$$f(0) = c > 0 :$$
$$V = \pi b^2 c + \int_c^{f(b)} \pi(b^2 - g(y)^2)\, dy$$